SUCTION CONTROL VALVE [SKYACTIV-D 1.5]

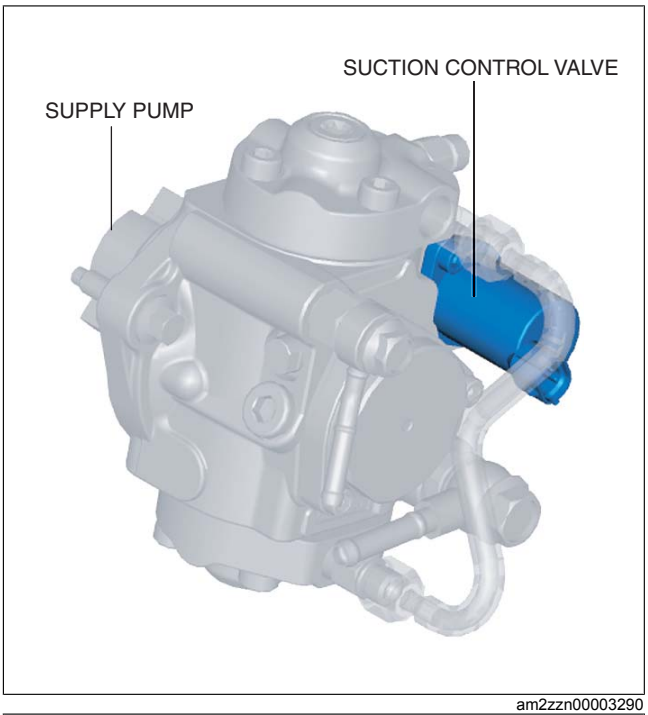
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Purpose, Function

- By controlling the supply pump intake amount, the suction control valve controls the pressure pump feed amount and the pressure inside the fuel distribution pipe.

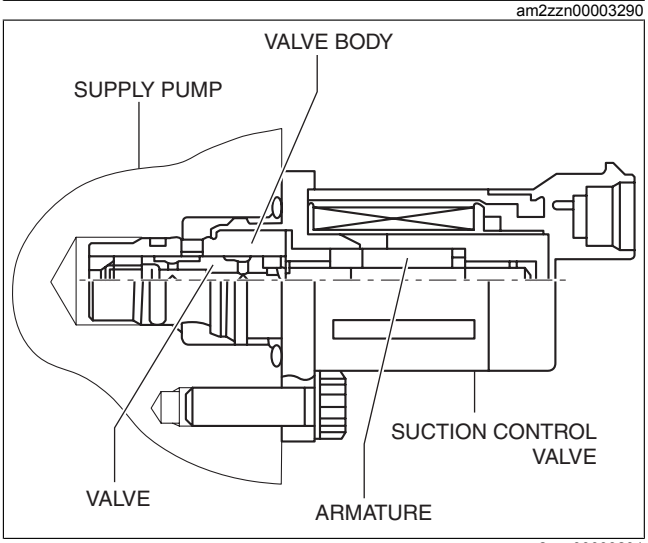
Construction

- The suction control valve is located in the supply pump.



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- The suction control valve consists of the valve, valve body, and armature.



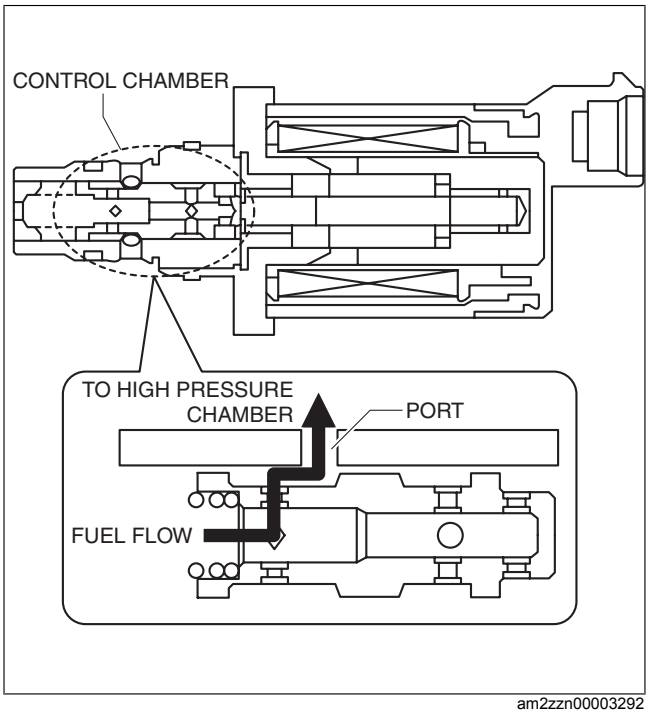
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Operation

- The solenoid drive current is duty controlled and the intake amount is adjusted by the change in the valve opening angle. When the solenoid drive current is small, the intake volume increases, and when it is large the intake volume decreases.

De-energized (open)

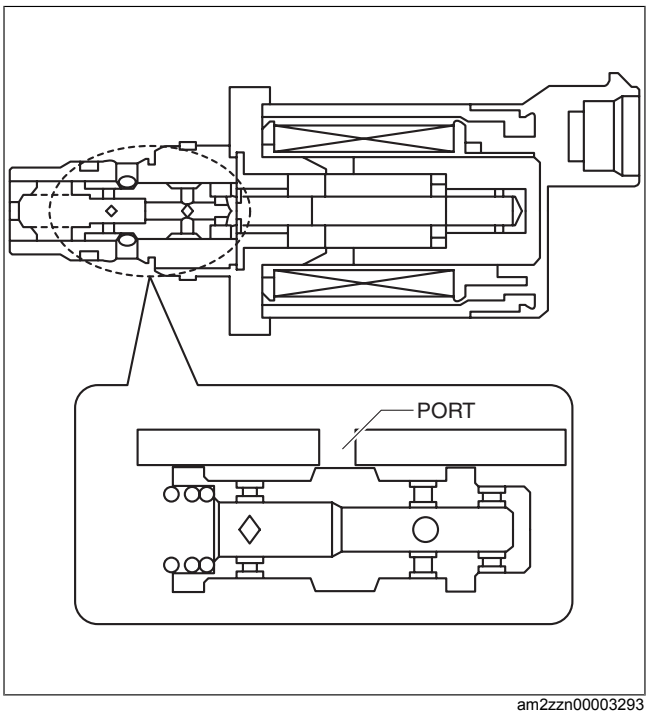
- The solenoid inner port is opened and fuel is sent into the control chamber.
- The port in the valve opens and sends fuel to the high pressure chamber.



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Energized (closed)

- The solenoid inner port is closed.



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Fail-safe

DTC No.	Fail-safe function
P009E:00	• Limits the engine torque.
P0089:00	• Limits the upper limit of the engine speed. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)

DTC No.	Fail-safe function
P062A:00	• Limits the engine torque. • Limits the upper limit of the engine speed. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P0087:00	• Inhibits the EGR control. • Increase the idle speed. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P0088:00	• Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)